ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	List the triples $a \le b \le c$ with $a + b > c$.
2	**(d)** 40°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
3	**(b)** 90°	Reason with the angle sum and exterior angle property.
4	**(a)** 80°	Find x from the angle sum, then compare the angles.
5	**(c)** no triangle can be formed	Check the sum of the two shorter sides.
6	**(d)** 10	Base $= P - 2s$.
7	**(a)** 30°	They add up to 90°.
8	**(b)** 3	List a, a, b with $b = P - 2a$ and $b < 2a$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	3 + 4 = 7 < 8, so the rule fails.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Two right angles would make 180 degrees with nothing left for the third angle. Both true, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Check where the equal parts lie in each triangle; No: SSA: the equal angle is not between the equal sides, so this is not a congruence rule (1); A different triangle with the same given parts can be drawn (1).	1+1
12	$\angle ACB = 180^\circ - 115^\circ = 65^\circ$; $\angle B = \angle ACB = 65^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 65^\circ = 50^\circ$ (1).	1+1
13	$b = P - 2a = 20 - 2a$ and $b < 2a$ so $a > 5$; Also $b > 0$ so $a < 10$ (1); $6, 6, 8$; $7, 7, 6$; $8, 8, 4$; $9, 9, 2$; Number of triangles $= 4$ (1).	1+1
14	Check where the equal parts lie in each triangle; No: AAA gives the same shape but not necessarily the same size (1); A different triangle with the same given parts can be drawn (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $4x + 60 + 3x + 100 = 230$ (1); $x = 10$; Exterior angles 100° and 130° (1); $\angle B = 80^\circ$, $\angle C = 50^\circ$ (1).	1+1+1
16	Base b, equal sides $b + 2$; $b + 2(b + 2) = 40$, so $3b = 36$ and $b = 12$ (1); Sides $14, 14, 12$ cm (1); $12 + 14 = 26 > 14$, so the triangle exists (1).	1+1+1
17	Third side between 13 and 29; Multiples of 5 in this range: $15, 20, 25$ (1); Largest third side 25 cm (1); Largest perimeter $= 8 + 21 + 25 = 54$ cm (1).	1+1+1
	OR: Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $5x + 50 + 4x + 62 = 220$ (1); $x = 12$; Exterior angles 110° and 110° (1); $\angle B = 70^\circ$, $\angle C = 70^\circ$ (1).	1+1+1
18	Base b, equal sides $b + 3$; $b + 2(b + 3) = 45$, so $3b = 39$ and $b = 13$ (1); Sides $16, 16, 13$ cm (1); $13 + 16 = 29 > 16$, so the triangle exists (1).	1+1+1
	OR: Third side between 13 and 25; Multiples of 4 in this range: $16, 20, 24$ (1); Largest third side 24 cm (1); Largest perimeter $= 6 + 19 + 24 = 49$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) No, as $4 + 5 = 9$ is not greater than 9 (1). (ii) Yes, as the sum of any two sides exceeds the third (1). (iii) Yes, $7 + 7 = 14 > 12$; it is isosceles (1). (iv) The third side is greater than $9 - 6 = 3$ cm and less than $9 + 6 = 15$ cm (1); whole numbers 4 to 14, i.e. 11 lengths (1).	1+1+1+2
20	Exterior angles add up to 360°; one part $= 360^\circ \div 15 = 24^\circ$ (1); Exterior angles 96°, 120°, 144° (1); Interior angles 84°, 60°, 36° (1); Check: 180° (1); The triangle is acute-angled (1).	1+1+1+1+1

	OR: Corresponding parts of congruent triangles are equal (1); $3x + 8 = 5x - 14$, so $2x = 22$ and $x = 11$ (1); $\angle A = 41^\circ$ (1); $2y + 5 = 3y - 3$, so $y = 8$ and $AB = 21$ cm (1); $\angle C = 180^\circ - 41^\circ - 30^\circ = 109^\circ = \angle F$ (1).	1+1+1+1+1

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
- In an isosceles triangle, taking the angles opposite the unequal side as equal.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 360°	Reason with the angle sum and exterior angle property.
2	**(c)** 70°	Find x from the angle sum, then compare the angles.
3	**(c)** a triangle can be formed	Check the sum of the two shorter sides.
4	**(b)** 14	Base $= P - 2s$.
5	**(b)** 36°	They add up to 90°.
6	**(b)** 4	List a, a, b with $b = P - 2a$ and $b < 2a$.
7	**(d)** 4	List the triples $a \le b \le c$ with $a + b > c$.
8	**(d)** 100°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
9	**(c)** A is true but R is false.	A is true. The exterior angle and the adjacent interior angle form a linear pair (sum 180 degrees), so R is false.
10	**(d)** A is false but R is true.	90 + 60 + 40 = 190, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle ACB = 180^\circ - 130^\circ = 50^\circ$; $\angle B = \angle ACB = 50^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 50^\circ = 80^\circ$ (1).	1+1
12	$b = P - 2a = 17 - 2a$ and $b < 2a$ so $a > 4.25$; Also $b > 0$ so $a < 8.5$ (1); $5, 5, 7$; $6, 6, 5$; $7, 7, 3$; $8, 8, 1$; Number of triangles $= 4$ (1).	1+1
13	Check where the equal parts lie in each triangle; Yes: SAS: the equal angle lies between the two equal sides (1); Matching parts of congruent triangles are equal (1).	1+1
14	$\angle ACB = 180^\circ - 145^\circ = 35^\circ$; $\angle B = \angle ACB = 35^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 35^\circ = 110^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $2x + 75 + 5x + 70 = 250$ (1); $x = 15$; Exterior angles 105° and 145° (1); $\angle B = 75^\circ$, $\angle C = 35^\circ$ (1).	1+1+1
16	Base b, equal sides $b + 1$; $b + 2(b + 1) = 38$, so $3b = 36$ and $b = 12$ (1); Sides $13, 13, 12$ cm (1); $12 + 13 = 25 > 13$, so the triangle exists (1).	1+1+1
17	Third side between 17 and 37; Multiples of 6 in this range: $18, 24, 30, 36$ (1); Largest third side 36 cm (1); Largest perimeter $= 10 + 27 + 36 = 73$ cm (1).	1+1+1
	OR: Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $6x + 62 + 2x + 114 = 240$ (1); $x = 8$; Exterior angles 110° and 130° (1); $\angle B = 70^\circ$, $\angle C = 50^\circ$ (1).	1+1+1
18	Base b, equal sides $b + 4$; $b + 2(b + 4) = 47$, so $3b = 39$ and $b = 13$ (1); Sides $17, 17, 13$ cm (1); $13 + 17 = 30 > 17$, so the triangle exists (1).	1+1+1
	OR: Third side between 13 and 31; Multiples of 3 in this range: $15, 18, 21, 24, 27, 30$ (1); Largest third side 30 cm (1); Largest perimeter $= 9 + 22 + 30 = 61$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\angle C = 180^\circ - 50^\circ - 70^\circ = 60^\circ$ (1). (ii) $\angle ACD = 180^\circ - 60^\circ = 120^\circ$ (1). (iii) $50^\circ + 70^\circ = 120^\circ$ (1). (iv) All angles are less than 90°, so it is acute-angled (1); the three angles are different, so the sides are different and it is scalene (1).	1+1+1+2
20	Exterior angles add up to 360°; one part $= 360^\circ \div 9 = 40^\circ$ (1); Exterior angles 80°, 120°, 160° (1); Interior angles 100°, 60°, 20° (1); Check: 180° (1); The triangle is obtuse-angled (1).	1+1+1+1+1
	OR: Corresponding parts of congruent triangles are equal (1); $3x + 6 = 5x - 24$, so $2x = 30$ and $x = 15$ (1); $\angle A = 51^\circ$ (1); $2y + 8 = 3y - 1$, so $y = 9$ and $AB = 26$ cm (1); $\angle C = 180^\circ - 51^\circ - 45^\circ = 84^\circ = \angle F$ (1).	1+1+1+1+1

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).

- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** no triangle can be formed	Check the sum of the two shorter sides.
2	**(d)** 11	Base $= P - 2s$.
3	**(d)** 40°	They add up to 90°.
4	**(a)** 4	List a, a, b with $b = P - 2a$ and $b < 2a$.
5	**(b)** 7	List the triples $a \le b \le c$ with $a + b > c$.
6	**(d)** 120°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
7	**(a)** two right angles	Reason with the angle sum and exterior angle property.
8	**(b)** 100°	Find x from the angle sum, then compare the angles.
9	**(a)** Both A and R are true and R is the correct explanation of A.	180 - 110 = 70.
10	**(a)** Both A and R are true and R is the correct explanation of A.	5 + 7 > 10, 5 + 10 > 7 and 7 + 10 > 5.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$b = P - 2a = 24 - 2a$ and $b < 2a$ so $a > 6$; Also $b > 0$ so $a < 12$ (1); $7, 7, 10$; $8, 8, 8$; $9, 9, 6$; $10, 10, 4$; $11, 11, 2$; Number of triangles $= 5$ (1).	1+1
12	Check where the equal parts lie in each triangle; Yes: ASA: two angles and the side between them are equal (1); Matching parts of congruent triangles are equal (1).	1+1
13	$\angle ACB = 180^\circ - 160^\circ = 20^\circ$; $\angle B = \angle ACB = 20^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 20^\circ = 140^\circ$ (1).	1+1
14	$b = P - 2a = 22 - 2a$ and $b < 2a$ so $a > 5.5$; Also $b > 0$ so $a < 11$ (1); $6, 6, 10$; $7, 7, 8$; $8, 8, 6$; $9, 9, 4$; $10, 10, 2$; Number of triangles $= 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $3x + 50 + 2x + 60 = 210$ (1); $x = 20$; Exterior angles 110° and 100° (1); $\angle B = 70^\circ$, $\angle C = 80^\circ$ (1).	1+1+1
16	Base b, equal sides $b + 5$; $b + 2(b + 5) = 52$, so $3b = 42$ and $b = 14$ (1); Sides $19, 19, 14$ cm (1); $14 + 19 = 33 > 19$, so the triangle exists (1).	1+1+1
17	Third side between 18 and 32; Multiples of 5 in this range: $20, 25, 30$ (1); Largest third side 30 cm (1); Largest perimeter $= 7 + 25 + 30 = 62$ cm (1).	1+1+1
	OR: Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $7x + 58 + 5x + 130 = 260$ (1); $x = 6$; Exterior angles 100° and 160° (1); $\angle B = 80^\circ$, $\angle C = 20^\circ$ (1).	1+1+1
18	Base b, equal sides $b + 2$; $b + 2(b + 2) = 34$, so $3b = 30$ and $b = 10$ (1); Sides $12, 12, 10$ cm (1); $10 + 12 = 22 > 12$, so the triangle exists (1).	1+1+1
	OR: Third side between 19 and 43; Multiples of 7 in this range: $21, 28, 35, 42$ (1); Largest third side 42 cm (1); Largest perimeter $= 12 + 31 + 42 = 85$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\dfrac{180^\circ - 40^\circ}{2} = 70^\circ$ (1). (ii) Acute-angled (1). (iii) $180^\circ - 70^\circ = 110^\circ$ (1). (iv) Perimeter $= 8 + 8 + 5 = 21$ m (1); no, because $8 + 8 = 16 < 17$ (1).	1+1+1+2
20	Exterior angles add up to 360°; one part $= 360^\circ \div 15 = 24^\circ$ (1); Exterior angles 72°, 120°, 168° (1); Interior angles 108°, 60°, 12° (1); Check: 180° (1); The triangle is obtuse-angled (1).	1+1+1+1+1
	OR: Corresponding parts of congruent triangles are equal (1); $3x + 14 = 5x - 22$, so $2x = 36$ and $x = 18$ (1); $\angle A = 68^\circ$ (1); $2y + 6 = 3y - 4$, so $y = 10$ and $AB = 26$ cm (1); $\angle C = 180^\circ - 68^\circ - 50^\circ = 62^\circ = \angle F$ (1).	1+1+1+1+1

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